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DEPARTMENT OF INFORMATION TECHNOLOGY

IT5302 – SOFTWARE ENGINEERING

PROJECT PROPOSAL

TOYAH

REVOLUTIONSING WATER MANAGEMENT FOR COMMUNITIES

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I. PROBLEM STATEMENT

'Ensure availability and sustainable management of water and sanitation for all.'

This is the 6th Sustainable Development Goal of the 17 established by the United Nations General Assembly in 2015. Water-related problems are some of the most commonly seen issues in society. Scarcity, contamination, lack of safe drinking water, and leakage are but some of the many problems that many people over the world face on a regular basis.

According to a 2020 report of WaterAid India, 1 in every 10 people do not have access to a safe drinking water source.

In India, water-borne diseases are estimated to pose an economic burden of approximately 600 million USD per year, as per a UNICEF update.

According to water.org, out of its population of 1.3 billion people, 91 million people (6% of the population) in India lack access to safely managed household sanitation facilities, and 746 million people (54%) lack access to safely managed household sanitation facilities.

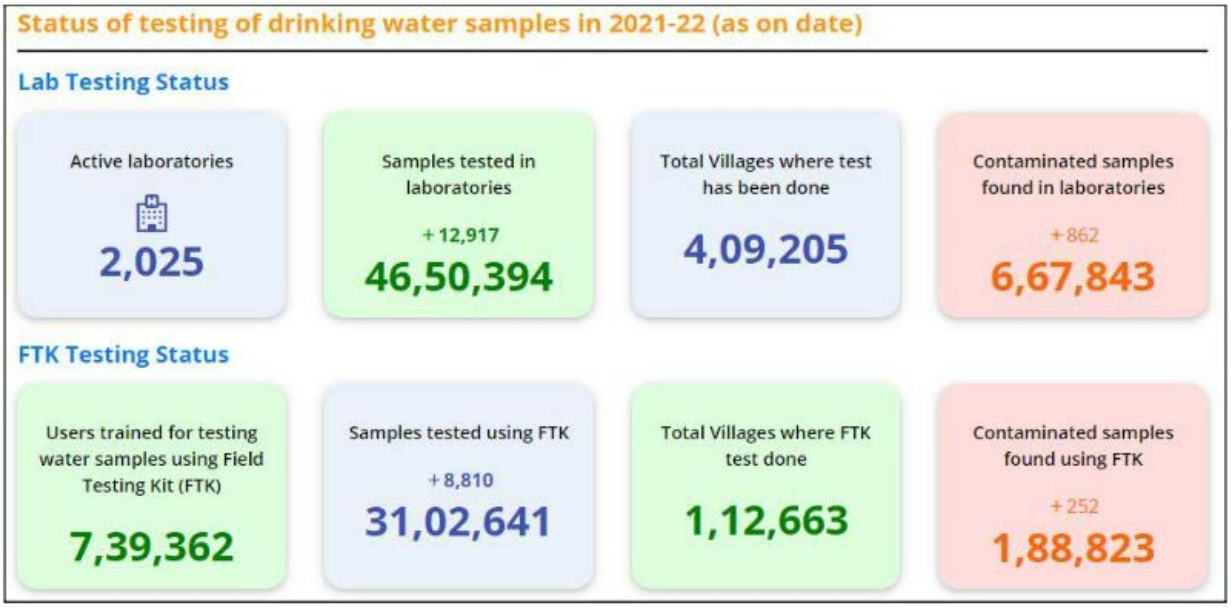
In times of cyclones and floods, water can enter into households, forcing people to evacuate to upper floors or shift to a new place entirely- and that is simply one of the many dozens of problems faced during such situations - is also not something that has been implemented in an effective manner.

So, existentially and even financially, all of these water problems are something that we should strive to get rid of. However, it is not possible without proper and reliable coordination between the affected people and the appropriate private and government organisations, NGOs and others who can effectively solve the issues.

II. EXISTING SOLUTIONS

In order to prevent the outbreak of water-borne diseases, the Government of India launched the National Rural Drinking Water Programme (NRDWP) in 2009, with the objective to provide every rural person with safe and adequate water for all purposes. This scheme was later subsumed under the Jal Jeevan Mission (JJM), which aims to provide potable water to every rural household by functional tap connection by the year 2024.

To ensure that the provided water is safe and meets acceptable quality standards, water testing is done, which currently involves field testing kits and collecting samples and testing them in laboratories. The status of country-wise availability of such laboratories and the samples tested during 2021-2022 as of 14 April 2022 is as below.



To capture and document the water level of selected 2-3 open wells in villages across the country, the Ministry of Rural Development and Panchayati Raj have jointly developed the JALDOOT Mobile Application. Gram Rojgar Sahayak (GRS) will measure the water level and document the same at the Central server using Jaldoot.

To address and deal with various problems including water-related problems, the Ministry of Housing and Urban Affairs developed Swachhata, a fourth generation complaint redressal mobile and web platform. It is an initiative of the Swachh Bharat Mission. It is a quantum leap in how complaints and grievances are being redressed by Municipal Corporations in India. This solution is for all the 4041 towns and cities of India. The core of this application is to use citizen participation and civic engagement to help resolve the Swachh Bharat complaints.

Several startups have been founded to deal with various water-related issues. Some of the most notable ones are OCEO Water, Kritsnam Technologies, WASU, and NeerAssure. All of these startups take up specific issues and provide innovative and simple ways to tackle and solve them.

In order to help improve groundwater management, which is one of the most important sources for irrigation, the World Bank is helping to support the government's national groundwater program, the Ataj Bhujal Yojana. Implemented in 8220 gram panchayats across seven Indian states, this is the world's largest community-led groundwater management program. Since groundwater conservation lies in the hands of hundreds of millions of individuals and communities, the program is helping villages understand their water availability and usage patterns so that they can budget their water use accordingly.

III. FLAWS WITH EXISTING SOLUTIONS

Some of the most important issues that are clearly seen in the aforementioned initiatives, platforms, startups or the like are explained below:

The solutions at hand are far too generalised to provide the necessary focus on problems associated with water. While applications like Swachhata do provide a meaningful way for people to share complaints and get them resolved as quickly as possible depending on the scale of the problem, it is restricted to the ability of the government and has a deeper focus on the more basic, regularly seen problems. In addition to this, due to the nature of the platform, immediate response is not really seen as often as one would generally expect it to be. Improvements are consistently being made, but again, they are not as fast as required.

As per a news report from The Times of India, in 2018, nearly 85% of the complaints posted on the Swachhata application were yet to be attended by the Trichy Corporation. Similar cases have been seen in many areas across the country.

Many of the initiatives and startup applications that have been established so far with respect to water management only focus on one or two specific water-related issues. They do undoubtedly tackle these problems in a very effective manner, but the fact remains that they do not provide the best solution cover. It is also difficult for these startups to reach as many people as they can without proper funding and support. Most do cross these hurdles and go on to become successful, while most do not. But again, as mentioned before, they don’t cover all aspects of a specific problem, and instead choose to focus on a subset of an existing specificity.

As we can clearly see, the existing solutions are either too generalised or too specific. So far, a platform dedicated to managing any and all water-related issues hasn’t been seen at a significant level. Such a platform is necessary since, as mentioned in the beginning, water-related issues are some of the most commonly seen problems in various parts of society.

IV. OUR SOLUTION

Clean and accessible water is a fundamental right for every human being. Our application, 'Toyah' (a Sanskrit word which means ‘water’), aims to revolutionize the way communities address water-related concerns, not limited to access or contamination or stagnation of water, but covering every possible aspect. This mobile application uses social media data shared by people and cutting-edge technology to crowdsource, analyse, and visualise water-related problems within communities.

The use of social media in disaster and crisis management is rapidly growing. In particular, geo-referenced images shared by citizens can provide situational awareness to emergency responders, as well as assist with financial loss assessment, giving information that would otherwise be very hard to collect through conventional sensors or remote sensing products. Our aim is to apply this principle to our specific concern of water-related problems, hoping to understand and help with any and all issues that communities are facing.

General Objectives

1. Effortless Reporting

Develop a user-friendly platform for community members to report water-related concerns seamlessly.

2. Integrated Satellite Data

Incorporate satellite data to provide an initial dataset, offering insights into local water bodies, infrastructure, and ecological conditions.

3. Advanced Pattern Analysis

Utilize data analytics to identify patterns and trends in water-related issues, enabling proactive solutions and sustainable water planning.

4. Transparency and Public Access

Ensure open access to collected data, fostering shared responsibility and accountability among individuals, non-profit organizations, and government entities.

5. Interactive Mapping Technology

Integrate advanced mapping technology to visually present reported issues on an interactive map, aiding community members in identifying and tracking areas of concern.

6. Community Engagement Hub

Establish 'Toyah' as a community-driven initiative, encouraging users to actively contribute reports, collaborate with non-profit organizations, and work together to address water-related problems effectively.

‘Toyah’ focuses on developing a mobile application equipped with features for reporting, satellite data integration, pattern analysis, transparency, and community engagement. It aims to be a balance of the existing solutions available for solving water-related problems, which, as mentioned before, are either too generalised or too specific.

V. IMPLEMENTATION TECHNIQUES

Methodology

1. User Interface Design

Develop an intuitive and user-friendly interface for easy reporting by community members.

2. Satellite Data Integration

Collaborate with satellite data providers to incorporate real-time data into the application.

3. Algorithm Development

Create algorithms for advanced pattern analysis and categorization of water-related issues.

4. Mapping Technology Integration

Utilize advanced mapping technology to visualize reported issues on an interactive map.

5. Community Engagement Strategies

Implement strategies to encourage active participation, collaboration, and a sense of shared responsibility among users.

Timeline

The project will be completed in approximately 4 months with key milestones including UI design completion, algorithm development, data integration, testing, and app launch.

App Walkthrough and Wire Framing

VI. CONCLUSION

The ultimate mission of ‘Toyah’ is to provide a natural, intuitive and easy to use method for its users to share their water-related concerns in an efficient manner. Since this application is being created as a balance of existing solutions, it has the potential to also act as a bridge in the future, connecting the very specific implementations and the general solutions.

As mentioned before, Sustainable Development Goal 6 of the 17 Sustainable Development Goals established by the United Nations is about ‘clean water and sanitation for all’. It is clearly seen that ‘Toyah’ not only aims to move closer towards the goal, but also cover every aspect, every path, every problem in whose roots lie the one single word: water.

With proper and consistent development, strategies to improve user interaction and engagement, and discussions to add new dimensions to the application in order to take it to the next level, ‘Toyah’ can truly become a landmark achievement in solving water-related issues for people everywhere and anywhere.